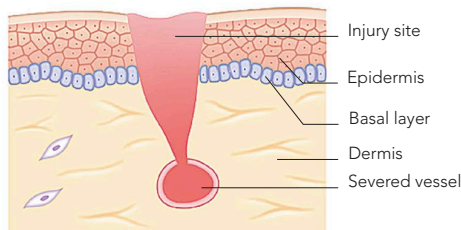


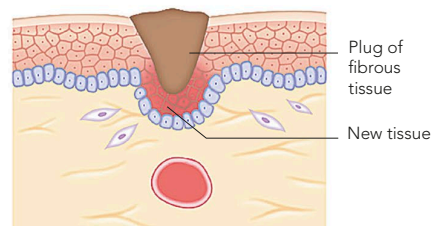
SKIN REPAIR

If the skin surface is breached, contents leak from damaged cells and stimulate the repair process. Platelets in the blood and the blood-clotting protein fibrinogen together form a mesh of fibres that traps red cells as the beginning of a clot.



1 INJURY

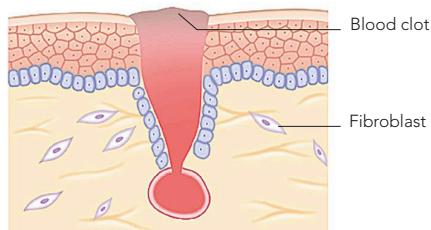
The wound breaks open cells and releases their contents. These components attract various defence and repair cells.



3 PLUGGING

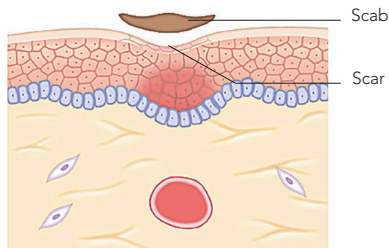
Fibroblasts produce a plug of fibrous tissue within the clot, which contracts and shrinks. New tissue begins to form beneath.

Tissue-forming fibroblast cells collect in the area, as do white cells called neutrophils, which ingest cell debris and foreign matter such as dirt and germs. The clot gradually hardens and expels fluid to become a scab, as the skin heals.



2 CLOTTING

Blood seeps from the vessel and forms a clot. Fibroblasts multiply and migrate to the damaged area.

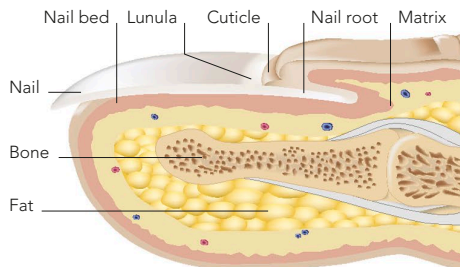


4 SCABBING

The plug hardens and dries into a scab, which eventually detaches. A scar may remain, but usually fades with time.

NAIL STRUCTURE

Fingernails and toenails are hard plates made of a tough protein called keratin. Growth takes place under a fold of flesh (cuticle) at the nail base. The nail matrix adds keratinized cells to the nail root, and the whole nail is continuously pushed forward along the nail bed towards its free edge. Most nails grow about 0.5mm ($\frac{1}{50}$ in) each week, with fingernails lengthening faster than toenails.



CROSS-SECTION THROUGH
NAIL AND FINGER